Sensors

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***Abstract*—This report provides an overview of the sensors required for Formula Student Driverless vehicles. It breaks down the core functionality, specific team use cases, and procurement strategies for various sensors.**

Keywords—Sensors, Driverless, IMU, LIDAR, GNSS

I. Introduction

The report breaks down the functionality, integration, troubleshooting, and Indian procurement strategies for perception sensors (LiDAR, Cameras), state estimation (IMU, INS, Optical, WSS), and actuation (Steer/Brake-by-wire).

II. Exteroceptive Perception (Seeing the Track)

A. LiDAR (Light Detection and Ranging)

1) Working: LiDAR operates on the active Time-of-Flight (ToF) principle. The sensor emits a near-infrared laser pulse (typically 905 nm). When the pulse reflects off a traffic cone, photons return to an avalanche photodiode. The distance to the cone is calculated using $D = (c \times \Delta t)/2$, where c is the speed of light and Δt is the round-trip time.

a) Mechanical Spinning LiDAR: Uses a physical motor to spin a vertical array of laser diodes across a full 360° horizontal Field of View (FOV). While providing total surround vision, the vertical resolution (the gap between laser rings) is often sparse.

b) Hybrid Solid-State / Directional LiDAR: Uses internal micro-prisms or MEMS mirrors instead of large spinning parts. This directs an ultra-dense, non-repetitive laser pattern primarily forward, yielding a significantly higher point density capable of resolving small objects at much greater distances.

2) Integration in the Car: LiDARs are optimally mounted directly above the front wing. This eliminates the "blind spot" caused by the car's nosecone, allowing the sensor to detect cones as close as 0.1 meters during tight Skidpad and Hairpin maneuvers.

3) Common Errors & How to Solve Them:

a) Motion Distortion (Smeared Point Clouds):

A 10 Hz scan takes 100ms. At racing speeds, the car translates significantly during one sweep, causing the resulting cone clusters to stretch or duplicate.

To solve this we perform "Point Cloud De-skewing" Feed high-frequency (1,000 Hz) IMU data into the perception pipeline to mathematically transform every individual laser return to the vehicle's position at the end of the scan.

b) Sparse Hits on Distant Cones

A standard FS cone is only 325mm tall. Mechanical LiDARs with large vertical gaps (e.g., 2.0° between rings) can shoot directly over and under a distant cone.

Solution is to rely on dense solid-state LiDARs or fuse the sparse 3D data with 2D high-resolution camera bounding boxes.

4) Specific Hardware & Indian Sourcing:

a) Livox Mid-360:

A hybrid solid-state LiDAR. Generates 200,000 points/sec with a 360° x 59° FOV. Includes a built-in IMU. Costs ₹78,000 - ₹85,000 via B2B importers on IndiaMART (e.g., Corebolt LLP, Shiva Enterprises).



Fig. 1. Livox Mid-360.

b) Hesai Pandar40P

A legacy mechanical spinning LiDAR featuring 40 channels. Excellent 360° reliability but heavier and costlier. Cost typically exceeds ₹3,00,000. Must be sourced directly via Hesai distributors or industrial B2B channels.



Fig. 2. Hesai Pandar40P.

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B. Cameras (Vision Pipelines)

1) Working:

Vision systems use CMOS sensors to convert incoming photons into electrical charge via the photoelectric effect and thereby detect color-coded cones. They operate at higher frequencies, typically 30 Hz to 60 Hz, to feed Convolutional Neural Networks (CNNs) like YOLO.

a) Monocular Cameras: A single-lens setup utilizing the pinhole projection model. Because depth cannot be directly measured from a 2D image, the software must use planar homography (assuming the ground is perfectly flat) or bounding-box height regression to estimate distance.

b) Stereo Cameras: Utilizes two lenses separated by a rigid horizontal baseline. By matching pixels across the left and right images (epipolar geometry), it calculates disparity to directly triangulate 3D depth. Stereo cameras utilize epipolar geometry: two lenses separated by a fixed baseline distance (b) capture synchronized frames. By finding a cone in both images, the pixel displacement (disparity, d) is measured. Depth (Z) is extracted via $Z = (f \times b)/d$, where f is the focal length. Because disparity is in the denominator, depth error grows quadratically with distance. Frequencies range from 30 Hz to 60 Hz.

2) Integration in the Car: Mounted at the highest point of the vehicle (the main roll hoop) to ensure a wide, unobstructed view of the track boundaries and distant corners.

3) Common Errors & How to Solve Them:

a) Rolling Shutter Warp: Standard sensors expose pixel rows sequentially. Chassis vibrations at racing speeds skew the image diagonally, ruining YOLO detection accuracy. Solution is to mandate the use of Global Shutter sensors, which expose the entire pixel array at the exact same microsecond, freezing motion perfectly.

b) Quadratic Depth Degradation. In stereo vision, depth uncertainty grows quadratically with distance. For example, a stereo camera is highly accurate at 5 meters but highly inaccurate at 25 meters. Solution is to use cameras for short-range dense mapping and color classification, but fuse the data with LiDAR for long-range distance accuracy.

4) Specific Hardware & Indian Sourcing:

a) Stereolabs ZED 2i: An active stereo camera with a 120mm baseline, global shutter sensors, and a built-in IMU, communicating via USB 3.0. Priced at ~₹73,000 from domestic suppliers like MG Super Labs (mgsl.in) or Fab.to.Lab.



Fig. 3. Stereolabs ZED 2i.

III. Proprioceptive State Estimation (Feeling the Car's Movement)

A. Inertial Measurement Units (IMU) and INS

1) Working: Modern Inertial Navigation Systems (INS) rely on Micro-Electro-Mechanical Systems (MEMS).

a) Accelerometers (MEMS Capacitive): Inside the die, a microscopic proof mass is suspended by polysilicon springs. Linear acceleration forces the mass to displace against the springs due to inertia (F=ma). This displacement changes the distance (d) between charged capacitor plates, altering capacitance ($C = \varepsilon A/d$). This shift translates into a proportional voltage representing linear specific force.

b) Gyroscopes (Coriolis Effect): A resonant driving mass is electrostatically oscillated at a high frequency (e.g., 20 kHz). When the vehicle rotates, a Coriolis force ($F_c = -2m(\omega \times v)$) pushes the mass perpendicularly to both the drive direction and rotation axis. This perpendicular deflection is measured capacitively to yield precise pitch, roll, and yaw rates.

c) Magnetometers (Ambient Magnetic Flux): Magnetometers function as digital compasses by measuring the Earth's ambient magnetic flux density to establish an absolute heading (yaw) reference. Internally, they often use magneto-resistive materials or Hall-effect elements where the Earth's weak magnetic field alters the flow of electrons in a semiconductor, changing its electrical resistance. This measurable change allows the sensor to calculate the vehicle's orientation relative to magnetic North.

2) Integration in the Car: Must be mounted exactly at the vehicle's Center of Gravity (CoG). If placed far away, the rotation of the car induces a physical centripetal acceleration that throws off the accelerometer. If you can't mount it at the CoG, you must write math in your Extended Kalman Filter (EKF) to subtract this effect.

3) Common Errors & How to Solve Them:

a) High-Frequency Vibration Aliasing: A stiff carbon fiber spaceframe transmits severe motor and road vibrations into the MEMS springs. The sensor can vibrate at resonant frequencies, mathematically rectifying zero-mean noise into a constant, fake acceleration. Solution is to mount the IMU on specialized elastomer damping plates to physically absorb and low-pass filter the chassis vibrations before they hit the silicon.

b) Magnetometer Failure: The 400V-600V tractive system creates a massive, swirling electromagnetic field (EMI) around the car, rendering magnetic compasses utterly useless for heading estimation. Solution is to disable the IMU's magnetometer. Rely on a Dual-Antenna RTK-GNSS (which measures the phase difference of satellite waves hitting two antennas to calculate a perfect heading) or SLAM algorithms.

4) Specific Hardware & Indian Sourcing:

a) VectorNav VN-200: A miniature, high-performance INS combining a MEMS IMU with a GPS receiver, outputting fused attitude/position at 400 Hz. Has high commercial cost. Best pursued via direct academic sponsorship from VectorNav.



Fig. 4. VectorNav VN-200.

b) Bosch Motorsport MM7.10 (IMU): 6-axis motorsport-grade IMU operating over CAN, incredibly resistant to vibrations. Can be sourced at around ~€940 via European dealers like XTRA Motorsport. Must use university DSIR exemption to waive ~49% customs duties.



Fig. 5. Bosch Motorsport MM7.10.

B. Wheel Speed Sensors (WSS)

1) Working: Wheel speed sensors operate at 100 Hz to 500 Hz to measure wheel angular velocity. Modern FSD cars use the Active Hall-Effect.

a) The Hall Effect Physics: Inside the sensor is a semiconductor plate with a steady electrical current flowing through it, backed by a permanent magnet. The magnetic field pushes the flowing electrons to one side of the plate, creating a baseline voltage. The sensor is mounted facing a metal gear (tone ring) on the wheel hub. As a steel gear tooth rotates directly in front of the sensor, it pulls and bends the permanent magnet's field lines. This changes how hard the electrons are pushed sideways in the semiconductor, changing the voltage. The sensor's internal circuitry converts these tiny voltage ripples into a clean, digital square wave that the ECU counts to measure RPM.

2) Integration in the Car: Mounted directly at the four wheel hubs facing the brake disc crenellations, or pulled virtually from the electric motor's internal shaft resolvers (if using in-hub motors).

3) Common Errors & How to Solve Them:

a) 4WD Traction Observability Dilemma: In a RWD car, the front wheels roll freely, giving you perfect ground speed. In a 4WD electric car, when you command maximum torque, all four tires slip longitudinally against the asphalt (slip ratio). The wheels are spinning faster than the car is moving. If the Kalman Filter trusts the wheel speed, it will think the car is flying, ruining your map. Solution is that wheel speed cannot be trusted as the ground truth velocity during acceleration or braking. It must be fused with Optical Ground Speed sensors or visual odometry.

4) Specific Hardware & Indian Sourcing:

a) ZF (Cherry) GS100502: A rugged, IP67-rated active Hall-effect gear-tooth sensor working up to 15 kHz. Can be sourced at ~₹3,300 per unit from authorized distributors like Element14 India or Mouser India.

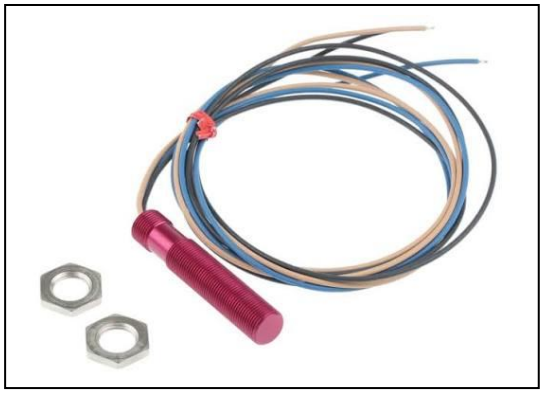


Fig. 6. ZF (Cherry) GS100502.

C. Optical Ground Speed Sensors

1) Working: These sensors act exactly like an incredibly powerful optical computer mouse, calculating the true, slip-free velocity of the vehicle over the road at 250 Hz to 1,000 Hz.

a) Spatial Surface Correlation: The sensor shines an intense LED or laser onto the asphalt. The rough micro-texture of the track scatters this light, which is captured by an internal photodiode array (a microscopic grid of light detectors). As the car moves, the optical texture of the road "slides" across the photodiode grid. The internal digital signal processor calculates the spatial frequency (how fast the light-dark patterns move from one pixel to the next) across both the X and Y axes. By performing this calculation thousands of times per second, it outputs the true longitudinal velocity and true lateral sliding velocity. From this, the true vehicle sideslip angle ($=\tan^{-1}$(lateral sliding velocity/true longitudinal velocity)) is directly calculated.

2) Integration in the Car: Mounted flush underneath the rear diffuser or sidepod undertray, pointing perpendicular to the track.

3) Common Errors & How to Solve Them:

a) Focal Length Dropouts: Massive aero downforce or severe pitch under heavy braking compresses the suspension. If the sensor gets too close or too far from the asphalt, the image blurs, and the spatial correlation algorithm fails. Solution is to mount the sensor near the rear axle (where pitch translation is minimized) and tune for ultra-stiff suspension setups.

4) Specific Hardware & Indian Sourcing:

a) Kistler Correvit SFII / Sensoric OMS Race: The absolute gold standard for measuring exact vehicle slip angles. Sourcing is far too expensive for retail purchase. Indian teams must submit technical proposals for academic loan sponsorships directly to Kistler Instrument India or Sensoric Solutions.

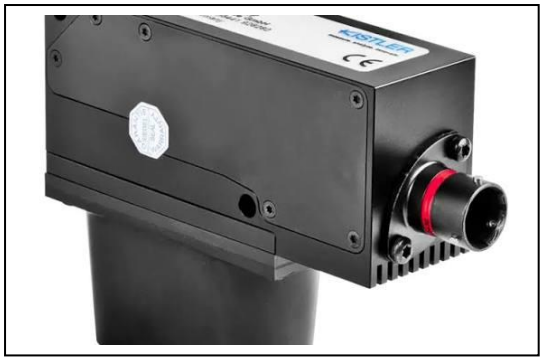


Fig. 7. Kistler Correvit SFII.

D. Global Navigation Satellite Systems (RTK-GNSS)

1) Working: Standard GPS is only accurate to 2-3 meters, which is useless on a tight 3-meter wide slalom track. FSD vehicles use Real-Time Kinematic (RTK) GNSS operating at 10 Hz - 20 Hz. RTK resolves the "integer cycle ambiguity" of the satellite carrier wave using corrections from a local base station, achieving centimeter-level accuracy. By using a Dual-Antenna setup with a known distance between them, the system uses carrier-phase interferometry to calculate a perfect, drift-free vehicle heading (yaw) even when the car is completely stationary at the start line.

2) Integration in the Car: Antennas mounted high on the chassis (usually main roll hoop or rear wing) with a clear view of the sky, separated by a rigid, fixed baseline distance.

3) Common Errors & How to Solve Them:

a) Multipath Reflections & Dropouts: Metallic grandstands or trees bounce the satellite signals. If the car relies on GPS to build its global map and the signal drops out, the map distorts, and the car crashes into the cones. Solution is to use Simultaneous localization and mapping. Never rely purely on GNSS for local cone tracking. Use GNSS as a reference, but rely on LiDAR/Camera SLAM (Simultaneous Localization and Mapping) to map the cones relative to each other.

4) Specific Hardware & Indian Sourcing: SBG Systems Ellipse-N / NovAtel PwrPak7D: Industry-standard dual-antenna RTK-GNSS. Sourcinging via standard retail sources is severely expensive. Indian teams acquire these via official academic sponsorships directly from SBG Systems.



Fig. 8. NovAtel PwrPak7D.

IV. Actuation and Safety Verification (Steer-by-Wire & Brake-by-Wire)

A. Steering Angle Sensors (SAS)

1) Working: Operating at 100 Hz to 1,000 Hz, these absolute magnetic rotary encoders read the physical angle of the steering rack. A diametrically magnetized magnet rotates above an IC. The IC contains an array of Hall sensors that measure the sine and cosine of the magnetic field strength, calculating the absolute shaft angle with sub-degree resolution.

2) Integration in the car: The Steering Angle Sensor is mounted directly onto the steering column shaft or bolted to the steering rack's pinion gear, physically adjacent to the autonomous steering servo motor.

3) Common Errors & How to Solve Them:

a) Steering Hysteresis (Backlash): Mechanical play between the steering pinion gear and the rack means the motor turns a few degrees before the wheels move.Solution is that the controller must use feed-forward algorithms to "punch" through the mechanical deadzone, guided by high-resolution SAS feedback.

4) Specific Hardware & Indian Sourcing:

a) Infineon AS5048A (IC) / Bosch LWS 7.3.6 (CAN Unit): 14-bit contactless magnetic rotary encoders. Sourced via Bare AS5048 chips (~₹800) or evaluation boards (~₹1,700) on Mouser India or DigiKey India.



Fig. 9. Bosch LWS 7.3.6.

B. Hydraulic Brake Pressure Transducers (Safety Critical ASB/EBS)

1) Working: Operating at 100 Hz to 500 Hz, these sensors use piezoresistive physics to measure hydraulic braking force. When brake fluid pushes against a micro-machined diaphragm inside the sensor, it bends. Piezoresistive materials attached to this diaphragm stretch, increasing their electrical resistance. Arranged in a Wheatstone bridge circuit, this unbalances the circuit and outputs a precise 0-5V signal directly proportional to the bar pressure of the brake fluid.

2) Integration in the Car: Plumbed directly into the M10x1 hydraulic ports of the front and rear brake master cylinders.

3) Common Errors & How to Solve Them:

a) ASB (Autonomous System Brake) Safety Failure: Emergency Brake System (EBS) must stop the car instantly if a fault occurs. If air entrains in the brake lines, hydraulic

pressure will not build. Solution is automated pre-start check. Before entering autonomous driving mode, the vehicle controller actuates the EBS valves and uses the Brake Pressure Transducers to verify that hydraulic pressure rapidly builds in both the front and rear redundant circuits. If the sensors do not register pressure, the car is hard-locked from driving.

4) Specific Hardware & Indian Sourcing:

a) Bosch Motorsport PSS-260 (0-260 bar) / PST-F 2: Rugged, motorsport-grade piezoresistive pressure transducers. Can be sourced at ~€80 from XTRA Motorsport. Indian teams must route this through their university's Stores and Purchase office using a valid DSIR Customs Duty Exemption Certificate to waive the crippling 49.2% import tax stack on automotive sensors.



Fig. 10. Bosch Motorsport PSS-260.

V. Extended Kalman Filter (EKF) and Sensor Fusion

A. Working (Algorithmic Principles):

Unlike a physical hardware sensor, the Extended Kalman Filter (EKF) is a recursive mathematical algorithm that acts as the "brain" of the state estimation layer. Because all sensors have noise and run at different frequencies, the EKF statistically fuses them to estimate the true state of the vehicle. Specifically its position, heading, planar velocities, and yaw rate. It operates in a continuous two-step loop:

1) Prediction Step (High Frequency): The EKF uses the high-rate IMU data (100-1,000 Hz) and mathematical kinematic models to continuously predict where the car will be in the next millisecond.

2) Measurement Update Step (Low Frequency): When slower, absolute sensors provide data (like a 10 Hz LiDAR map update or 20 Hz RTK-GNSS position), the EKF calculates the error between its IMU-based prediction and the new absolute measurement. It then applies a "Kalman Gain" to correct the vehicle's estimated position. Because FSD vehicle dynamics are highly nonlinear, the Extended Kalman Filter linearizes these physics equations using Taylor series expansions (Jacobians) around the current estimate.

B. Integration in the Car (Software Architecture):

The EKF exists purely in software, housed within the main compute unit (e.g., NVIDIA Jetson). It acts as the central hub of the autonomous stack, subscribing to raw data from the IMU, RTK-GNSS, Steering Angle Sensor, and Wheel Speed Sensors, and publishing one clean, unified "Odometry" message to the path-tracking controller (like the Model Predictive Controller) so the car knows exactly how to steer.

C. Common Errors & How to Solve Them:

1) Sensor Latency and Out-of-Sequence Measurements: An IMU sends data instantly, but processing a 3D LiDAR point cloud takes about 50-100 milliseconds. If the EKF blindly fuses the LiDAR data when it arrives, it applies old data to the car's current position, causing the vehicle to severely oscillate or crash. Solution to this is that teams must implement a Delay-Compensated EKF. The software maintains a history buffer (ring buffer) of past states. When delayed LiDAR data arrives, the EKF mathematically "rolls back time" to when the laser actually fired, applies the correction, and fast-forwards the math back to the present.

2) Trusting Corrupted Data (Covariance Tuning): During aggressive 4WD acceleration, the wheels slip. If the EKF trusts the Wheel Speed Sensors, it will falsely estimate the car is moving faster than it is. Solution is Adaptive Covariance Matrix tuning. The EKF is programmed to actively monitor tire slip; when slip exceeds a safe threshold, the algorithm dynamically "inflates the covariance" (meaning it lowers its mathematical trust) of the wheel speed sensors and temporarily relies heavier on the Optical Ground Speed sensor or GNSS.

D. Specific Frameworks & Implementation:

Rather than writing complex matrix calculus from scratch in C++, top Formula Student teams generate their EKF C++ code using MATLAB/Simulink's Sensor Fusion and Tracking Toolbox. Alternatively, teams use ROS 2, which provides highly optimized, pre-written EKF nodes configured specifically for fusing 3D IMU, GPS, and odometry data in autonomous robots.

VI. Computational Hardware (Edge AI Accelerators)

A. Working (Computational Principles):

An FSD vehicle generates a massive throughput of sensor data. Standard microcontrollers (CPUs) process tasks sequentially and will instantly bottleneck under this load. Edge AI computers solve this using System-on-Module (SoM) architectures featuring thousands of GPU (Graphics Processing Unit) cores. These GPU cores use parallel processing (CUDA) to execute multiple math operations simultaneously. This is strictly required to run deep learning Convolutional Neural Networks (like YOLOv7 trained on the FSOCO dataset for cone color classification) alongside the heavy matrix algebra required for Euclidean LiDAR clustering and FastSLAM.

B. Integration in the Car (Hardware & Interfaces):

The compute unit acts as the "brain" and sits at the top of the software hierarchy, typically running the Robot Operating System (ROS 2). Physically, it is usually packaged inside a custom IP67-rated (waterproof) enclosure mounted behind the firewall or inside the sidepods. It interfaces directly with sensors via high-bandwidth ports: Gigabit Ethernet (for LiDAR), USB 3.0 (for cameras), and a CAN bus interface to send target steering and braking commands down to the low-level microcontrollers controlling the actuators.

C. Common Errors & How to Solve Them:

1) Thermal Throttling:

Running complex neural networks at full capacity generates immense heat. If the silicon exceeds its safe temperature limits, the GPU will automatically "throttle" (slow down) to prevent melting. This drops the perception pipeline's frame rate from 30 Hz to 5 Hz, creating massive latency that causes the car to overshoot corners. The solution is teams must discard standard desktop fans and design custom active cooling solutions.

D. Specific Hardware & Indian Sourcing:

a) NVIDIA Jetson AGX Orin: The industry standard for FSD, delivering up to 275 TOPS (Trillion Operations Per Second) of AI performance. Readily procured through authorized global distributors operating in India, such as Arrow Electronics or Mouser India, costs INR 4.0 Lakhs and INR 5.5 Lakhs per unit.



Fig. 11. NVIDIA Jetson AGX Orin.

VII. Conclusion

Designing an effective sensing architecture for Formula Student Driverless requires a deep understanding of physical sensor limits. By coupling robust solid-state LiDARs and global-shutter vision with high-frequency Hall-effect and inertial state estimation, teams can navigate the track with centimeter-level precision. Executing this architecture successfully demands strict adherence to optimized B2B sourcing and finding possible ways to cut down taxes and duties by sourcing from OEMs within the India.

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